

<i>Title:</i>	Revision of the Bachelor of Science in Chemical Engineering
<i>Sponsor:</i>	<i>Department of Chemical Engineering</i>
<i>Executive Summary:</i>	The prerequisite courses in the “Minor in Chemical Engineering” require updating to allow students to take: 1) CHEM 342 or CHE201 and 2) CS109 or CHE205. There is no change in the Total Hours for the Minor in Chemical Engineering
<i>Description:</i>	<i>Change the following two prerequisite course requirements for the Minor to:</i> <i>CHEM 342 or CHE 201</i> <i>CS109 or CHE205</i>
<i>Justification:</i>	- CHE 201 is a pre-requisite for the 300-level ChE courses and should be added to the prerequisite list. However, CHEM342 is sufficient for satisfying the CHE201 prerequisite and will allow LAS students to obtain a minor in CHE without the burden of an extra course. - CHE 205 is a pre-requisite for the 300-level ChE courses and should be added to the prerequisite list. However, CS109 is sufficient for satisfying the CHE205 prerequisite and will allow students who have taken CS109 to obtain a minor in CHE without the burden of an extra course.
<i>Catalog Statement:</i>	<i>See below</i>
<i>Minority Impact Statement:</i>	<i>N/A</i>
<i>Budgetary and Staff Implications:</i>	<i>N/A</i>
<i>Library Resource Implications:</i>	<i>N/A</i>
<i>Space Implications:</i>	<i>N/A</i>

<i>Unit (e.g. department) approval date:</i> <i>College (educational policy committee, faculty) approval dates:</i>	
<i>Contact Person:</i>	Alan Zdunek zdunek@uic.edu
<i>Proposed Effective Date/Term:</i>	<i>Example:</i> Fall, 2016

Current	Revised																						
BS in Chemical Engineering <i>Degree Requirements</i> To earn a Bachelor of Science in Chemical Engineering degree from UIC, students need to complete University, college, and department degree requirements. The Department of Chemical Engineering degree requirements are outlined below. Students should consult the <i>College of Engineering</i> section for additional degree requirements and college academic policies. BS in Chemical Engineering Degree Requirements Hours Nonengineering and General Education Requirements 74 Required in the College of Engineering 48 Technical Elective ^a 3 Electives outside the Major Rubric ^a 3 Total Hours—BS in Chemical Engineering 128 <i>a Students in the Biochemical Engineering Concentration take a minimum of 8 hours of electives and 130 hours for the degree; see below.</i> Nonengineering and General Education Requirements <table> <tr> <th>Courses</th><th>Hours</th></tr> <tr> <td>ENGL 160— Academic Writing I: Writing for Academic and Public Contexts</td><td>3</td></tr> <tr> <td>ENGL 161— Academic Writing II: Writing for Inquiry and Research</td><td>3</td></tr> <tr> <td>Exploring World Cultures course^a</td><td>3</td></tr> <tr> <td>Understanding the Creative Arts course^a</td><td>3</td></tr> <tr> <td>Understanding the Past course^a</td><td>3</td></tr> <tr> <td>Understanding the Individual and Society course^a</td><td>3</td></tr> <tr> <td>Understanding U.S. Society course^a</td><td>3</td></tr> <tr> <td>MATH 180—Calculus I^b</td><td>5</td></tr> <tr> <td>MATH 181—Calculus II^b</td><td>5</td></tr> <tr> <td>MATH 210—Calculus III^b</td><td>3</td></tr> </table>	Courses	Hours	ENGL 160— Academic Writing I: Writing for Academic and Public Contexts	3	ENGL 161— Academic Writing II: Writing for Inquiry and Research	3	Exploring World Cultures course ^a	3	Understanding the Creative Arts course ^a	3	Understanding the Past course ^a	3	Understanding the Individual and Society course ^a	3	Understanding U.S. Society course ^a	3	MATH 180—Calculus I ^b	5	MATH 181—Calculus II ^b	5	MATH 210—Calculus III ^b	3	BS in Chemical Engineering SAME SAME SAME
Courses	Hours																						
ENGL 160— Academic Writing I: Writing for Academic and Public Contexts	3																						
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MATH 180—Calculus I ^b	5																						
MATH 181—Calculus II ^b	5																						
MATH 210—Calculus III ^b	3																						

MATH 220—Introduction to Differential Equations	3	
PHYS 141—General Physics I (Mechanics) ^b	4	
PHYS 142—General Physics II (Electricity and Magnetism) ^b	4	
CHEM 112—General College Chemistry I ^b	5	
CHEM 114—General College Chemistry II ^b	5	
CHEM 222—Analytical Chemistry	4	
CHEM 232—Organic Chemistry I	4	
CHEM 233—Organic Chemistry Laboratory I	1	
CHEM 234—Organic Chemistry II	4	
CHEM 342—Physical Chemistry I	3	
CHEM 346—Physical Chemistry II	3	
Total Hours—Nonengineering and General Education Requirements	74	
a Students should consult the General Education section of the catalog for a list of approved courses in this category.		
b This course is approved for the Analyzing the Natural World General Education category.		
Required in the College of Engineering		
Courses	Hours	SAME
ENGR 100—Orientation ^a	0 ^a	
CHE 201—Introduction to Thermodynamics	3	
CHE 205—Computational Methods in Chem. Eng.	3	
CHE 210—Material and Energy Balances	4	
CHE 301—Chemical Engineering: Thermodynamics	3	
CHE 311—Transport Phenomena I	3	
CHE 312—Transport Phenomena II	3	
CHE 313—Transport Phenomena III	3	
CHE 321—Chemical Reaction Engineering	3	
CHE 341—Chemical Process Control	3	
CHE 381—Chemical Engineering Laboratory I	2	
CHE 382—Chemical Engineering Laboratory II	2	
CHE 396—Senior Design I	4	
CHE 397—Senior Design II	3	
CHE 499—Professional Development Seminar	0	
ECE 210—Electrical Circuit Analysis	3	
CME 260—Properties of Materials	3	
CS 109—C/C++ Programming for Engineers with MatLab	3	
Total Hours—Required in the College of Engineering	48	
a ENGR 100 is one-semester-hour course, but the hour does not count toward the total hours required for graduation.		
Technical Elective		SAME
Courses	Hours	
One technical elective to be chosen from the following list of design-oriented courses:^a	3	
CHE 392—Undergraduate Research (3) ^b		
CHE 410—Transport Phenomena (3)		

CHE 413—Introduction to Flow in Porous Media (3) CHE 421—Combustion Engineering (3) CHE 422—Biochemical Engineering (3) CHE 423—Catalytic Reaction Engineering (3) CHE 425 Nanotechnology Drug Deliver (3) CHE 431—Numerical Methods in Chemical Engineering (3) CHE 433 Process Simulation with Aspen Plus (3) CHE 438—Computational Molecular Modeling (3) CHE 440—Non-Newtonian Fluids (3) CHE 441—Computer Applications in Chemical Engineering (3) CHE 445—Mathematical Methods in Chemical Engineering (3) CHE 450—Air Pollution Engineering (4) CHE 456—Fundamentals and Design of Microelectronics Processes 930 CHE 494—Selected Topics in Chemical Engineering (3) Total Hours—Technical Elective	3
a Possible technical elective credit for a 400-level CHE course not listed above will require departmental approval by petition to the Undergraduate Committee. b An appropriate design-related research project may be selected with the approval of the Department of Chemical Engineering.	
Courses	Hours
Electives outside the CHE rubric	3
Total Hours—Electives outside the Major Rubric	3

<u>CHEM 352</u>	Introductory Biochemistry	
<u>CHEM 452</u>	Biochemistry I	
Total Hours	8-10	
^a Due to structure of the concentration and the prerequisites required for some of the courses, students in the concentration will be required to take a minimum of 130 semester hours for the degree.		
Sample Course Schedule		SAME
Freshman Year		
First Semester	Hours	
MATH 180—Calculus I	5	
CHEM 112—General College Chemistry I	5	
ENGL 160—Academic Writing I: Writing for Academic and Public Contexts	3	
CS 109—C/C++ Programming for Engineers with MatLab	3	
ENGR 100—Orientation ^a	1 ^a	
Total Hours	16	
Second Semester	Hours	SAME
MATH 181—Calculus II	5	
PHYS 141—General Physics I (Mechanics)	4	
ENGL 161—Academic Writing II: Writing for Inquiry and Research	3	
CHEM 114—General College Chemistry II	5	
Total Hours	17	
		SAME
Sophomore Year		
First Semester	Hours	
MATH 210—Calculus III	3	
PHYS 142—General Physics II (Electricity and Magnetism)	4	
CHEM 232—Organic Chemistry I	4	
CHE 201—Introduction to Thermodynamics	3	
General Education Core course	3	
Total Hours	17	
Second Semester	Hours	SAME
MATH 220—Introduction to Differential Equations	3	
CHEM 234—Organic Chemistry II	4	
CHEM 233—Organic Chemistry Lab I	1	
CHE 205—Computational Methods in Chem. Eng.	3	
CHE 210—Material and Energy Balances	4	
CME 260— Properties of Materials	3	

Total Hours	18	
Junior Year		
First Semester	Hours	SAME
ECE 210—Electrical Circuit Analysis	3	
CHE 301—ChE Thermodynamics	3	
CHE 311—Transport Phenomena I	3	
CHEM 222—Analytical Chemistry	4	
General Education Core course	3	
Total Hours	16	
Second Semester	Hours	SAME
CHEM 342—Physical Chemistry I	3	
CHE 312—Transport Phenomena II	3	
CHE 313—Transport Phenomena III	3	
CHE 321—Chemical Reaction Engineering	3	
General Education Core course	3	
Total Hours	15	
Senior Year		
First Semester	Hours	SAME
CHE 381—Chemical Engineering Laboratory I	2	
CHE 396—Senior Design I	4	
CHEM 346—Physical Chemistry II	3	
CHE Technical elective—Selected from CHE 410, 413, 421, 422, 423, 431, 438, 440, 441, 445, 450, 456, 494, or 392 (departmental approval is required for CHE 392)	3	
General Education Core Course	3	
Total Hours	15	
Second Semester	Hours	SAME
CHE 382—Chemical Engineering Laboratory II	2	
CHE 341—Chemical Process Control	3	
CHE 397—Senior Design II	3	
CHE 499—Professional Development Seminar	0	
Elective outside the major rubric	3	
General Education Core course	3	
Total Hours	14	

Minor in Chemical Engineering	Minor in Chemical Engineering
For the minor, 16–18 semester hours are required, excluding prerequisite courses. Students outside the Department of Chemical Engineering who wish to minor in Chemical Engineering must complete the following:	For the minor, 16–18 semester hours are required, excluding prerequisite courses. Students outside the Department of Chemical Engineering who wish to minor in Chemical Engineering must complete the following:
Prerequisite Courses	Prerequisite Courses
CHEM 112 General College Chemistry I 5 or CHEM 116 Honors General Chemistry I	CHEM 122 General College Chemistry I (Lecture) 4 CHEM 123 General College Chemistry I (Laboratory) 1 Or CHEM 116 Honors General Chemistry I 5
CHEM 342 Physical Chemistry I 3	CHEM 342 or CHE 201 Physical Chemistry I or Introduction to Thermodynamics 3
CS 109 C/C ++ Programming for Engineers with MatLab 3	CS 109 or CHE205 C/C++ Programming for Engineers with MatLab or Computational Methods in Chemical Engineering 3
MATH 180 Calculus I 5	MATH 180 Calculus I 4
MATH 181 Calculus II 5	MATH 181 Calculus II 4
MATH 210 Calculus III 3	
MATH 220 Introduction to Differential Equations 3	
PHYS 141 General Physics I (Mechanics) 4	
PHYS 142 General Physics II (Electricity and Magnetism) 4	
Total Hours 35	Total Hours 33
Required Courses	SAME
CHE 210 Material and Energy Balances 4	
CHE 301 Chemical Engineering Thermodynamics 3	
CHE 321 Chemical Reaction Engineering 3	
CHE 311 Transport Phenomena I 3-4 or ME 211 Fluid Mechanics I	
Select one of the following: 3-4	
CHE 312 Transport Phenomena II	
ME 321 Heat Transfer	
CHE 313 Transport Phenomena III	
Total Hours 16-18	